

Applied Machine Learning & Data Science Projects

Data Science 435 / 535 & Computer Science 449 / 549 | Fall 2026

Time & Location:

Mondays & Wednesdays 4:00pm - 5:30pm

Location: OES 130

Instructors:

Dr. Arko Barman (arko.barman@rice.edu)

Dr. Xinjie Lan (xl116@rice.edu)

Dr. Chad Shaw (cashaw@rice.edu)

Office Hours:

Dr. Arko Barman (Location = Duncan 1036 if $n \leq 2$ else Zoom) - By appointment

Dr. Xinjie Lan (Location = Maxfield 217) - By appointment

Dr. Chad Shaw - By appointment

Course Description:

In this project-based course, student teams will complete semester-long data science research or analysis projects selected from a variety of disciplines and industries. Students will also learn best practices in data science.

Course Overview:

This is a project course where student teams will complete semester-long data science research or analysis projects selected from a variety of disciplines and industries. Each student team will have a Rice instructor/faculty mentor, a PhD student (fellow) mentor, and a sponsor mentor. Student teams are required to meet weekly outside of class as a team without mentors as well as with their PhD fellow and sponsor mentors. During the semester, teams will work towards project milestones and present their progress in interim reports, oral presentations, and software checks. At the end of the semester, each team will produce a final project report with accompanying software and give a presentation on their work at the D2K Showcase. This course will also have a lecture component at the beginning of the semester where students will learn how to design a data science pipeline and best practices in data science with emphasis placed on computational and data scientific reproducibility as well as the fundamentals of project management.

Course Objectives & Outcomes:

In this course, students will engage in experiential learning where they learn best practices in data science by working on real-world data science projects. Students will learn how to wrangle, explore, and model their data as well as how to design a reproducible and reliable data science pipeline and validate their data science findings. Through these projects, students will also learn team-work,

leadership, project management, and communication skills. Students completing this course should be able to

1. Successfully complete a data science research or analysis project;
2. Successfully design and deploy a computationally and scientifically reproducible data analysis pipeline;
3. Work collaboratively on a team to complete a data science project, write reports, and give presentations;
4. Effectively communicate their findings to both a general and a technical audience;
5. Develop and manage a data science software repository intended for external use.

DSCI 435 / 535 vs. COMP 449 / 549:

Computer science students should enroll in COMP 449 or COMP 549, and all other students (including masters in data science students) should enroll in DSCI 435 or DSCI 535. Graduate students should enroll in the 500-level course and undergrads in the 400-level course. Graduate students are expected to help lead their team as well as provide major technical contributions to the team project.

Course Application & Project Placement:

This course requires an application for project placement and team assignment. The course application will ask students to highlight their data science background and experience as well as list their project preferences and interests. Students will be placed in teams for specific projects based on their background and skill set as well as their project preferences.

Lectures:

At the beginning of the semester, there will be a small number of lectures (some of them online) and discussions about the data science pipeline, how to design a pipeline, how to manage projects and organize meetings, best practices for software development, and using git.

Classes & Monday or Wednesday Teams:

After the first month of class (see schedule for details), teams will be divided into Monday or Wednesday teams. Students are only expected to attend class on their assigned day. All software checks, and reports will be due by 11.59pm on the team's assigned day. All presentations will be due by 3.55pm on the team's assigned day.

Coursework:

The major work of this course will be to conduct a data science research or analysis project. Each team will present their work through presentations at the D2K Showcase and through a final report with accompanying software due at the end of the semester. Throughout the semester, shorter interim reports, presentations, and software checks will be due to give each team feedback and guidance on how best to prepare their final report and presentation. Further details and instructions on the reports, presentations, and software checks as well as grading rubrics are given in a separate document.

Course Expectations:

Students are expected to work at least 10 hours a week on their project outside of class. Each week outside of class, teams are required to have one meeting with their mentors and at least one separate meeting with just team members. Students may conduct these team meetings over zoom or in person. Attendance at team meetings, lectures, team check-ins, and presentations is mandatory unless prior special arrangements have been made. Additionally, each team member is expected to contribute to all aspects of the data science project including technical data science contributions, coding, and preparing the written report and presentations.

Professional writing, presentation, and coding standards will be expected from students. Tutorials, lectures, and workshops will be conducted to help students learn and emulate these professional standards. Students are also encouraged to actively seek feedback from their fellow mentors and faculty mentors to improve their writing, presentation, and coding to be at par with professional standards.

AI Policy:

- Students are allowed to use AI tools to complete the assignments including report drafting, presentation, and software.
- Students are forbidden to plagiarize the work of AI tools. You will get **0 points** for any assignment if your submission is recognized as a complete or partial copy of the content generated from AI tools.
- AI tools can be used as an assistant only, but you must take the responsibility to review, understand, upgrade, and polish the content generated by AI tools, and demonstrate that you understand what AI tools generated and transfer AI generated content as your own knowledge and skills for completing the required assignments.
- Sponsor's data is not allowed to be uploaded to AI tools without sponsor's written authorization.

Grading Overview:

- Individual: 30%
 - Self and Peer evaluations: 10%
 - Individual project contributions: 10%
 - Class participation & Oral Examination: 10%
 - Bonus: 5% for regular participation in Socratic circle study
- Team: 70%
 - Interim Evaluations: 15%
 - Final Report: 25%
 - Final Presentation: 15%
 - Final Software: 15%

Grading rubrics as well as details on all coursework are given in a separate document. Note that students who do not contribute in any meaningful way to the team project will receive a failing grade for this course, regardless of the team grade.

Team Contracts:

All students in a team must discuss the terms of a team contract. They must sign the contract and submit it to the instructors within the end of the 3rd week of class. The terms of the contract are binding and the penalties for the violations of the contract will be decided upon by the team and finalized by the team in consensus, based on the recommendations of the instructor.

Computing Resources:

Computing resources and storage will be provided by Rice University CRC. Training on how to use these resources will also be provided. While the instructors will provide you guidance on how to use Rice storage systems, please note that all questions regarding computational usage must be directed to Rice University OIT. The instructors will not be able to respond to these queries.

For certain projects, the data, storage, and computing resources are provided by the sponsoring organization/company through their own cloud computing environments. In these cases, please consult your sponsor and your faculty mentor what data and analysis can be used by you outside these environments.

Study on Teaching & Learning:

This semester your instructors will be conducting a study on teaching and learning. The purpose of this study is to determine the factors that influence teaching effectiveness and learning. Your participation in this study will last for the duration of the current semester and will entail activities no different from the regular activities you would otherwise engage in as part of the course. If you would like more information, please do not hesitate to ask. If you would like to opt out of the study, please talk to the instructors as soon as possible.

Course Schedule:

- Red: all students need to attend class.
- Blue: holiday.
- Black: students only need to attend Monday or Wednesday class.

Week	Date	Class Time	Assignments
1	Aug-24	Course Intro + Project Intro Course & Project Application guidance (AB+CS+XL)	
1	Aug-26	Lecture: Data science pipeline lecture (AB) + Case Studies (AB+XL)	Module 1 Data science pipeline quiz Due 11.59am Course & Project Application Due 11.59pm
2	Aug-31	Lecture: Project Management (CS) + Meet teammates and fellow mentors	
2	Sep-2	Team Contracts + Meet Sponsors	Module 2 Project management quiz Due 11.59am
3	Sep-7	Labor Day	
3	Sep-9	Lecture: software development (AB) + Socratic circle.	Module 3 Software development quiz Due 11.59am
4	Sep-14	Lecture: AI tools for capstone (XL)	

	Sep-16	Lecture: Academic writing (CS)	
5	Sep-21 Sep-23	Best practices for slides (Dr. Tracy Volz)	
6	Sep-28 Sep-30	Initial Presentations	Initial Report + Initial Self & Peer Evaluations Due 11.59pm
7	Oct-5 Oct-7		
8	Oct-12 Oct-14	Fall Recess	
9	Oct-19 Oct-21	Software Review by instructors	
10	Oct-26 Oct-28	Midterm presentation	Midterm Report + Midterm Self & Peer Evaluations Due 11.59pm
11	Nov-2 Nov-4		Midterm Software Due 11.59pm
12	Nov-9 Nov-11		
13	Nov-16 Nov-18	Best practices for posters (Dr. Tracy Volz)	
14	Nov-23 Nov-25	Thanksgiving Break	
15	Nov 30 Dec-2	Final Presentation	
16	Week of Dec-7	D2K Showcase	
Final Exam Period	Dec-13		Final Report + Final Software + Final Self & Peer Evaluations Due

Accommodations:

The Americans with Disabilities Act (ADA; 1990) ensures that all qualified individuals have equal access to educational opportunities, regardless of disability. Students with a documented disability who require academic accommodations should contact the Student Disability Resource Center (SDRC; <https://sdr.c.rice.edu>) to submit appropriate documentation or confirm that their records are current. The SDRC is located in Allen Center 111, and can also be reached at sdr.c@rice.edu or x5841.

For additional information regarding access and accommodations for individuals with disabilities, please review Rice University Policy 402: <https://policy.rice.edu/402>.

Students with a documented disability requiring accommodations should speak with the instructors during the first two weeks of class.

Course change policy:

This syllabus is subject to change with reasonable advance notice by the instructors.